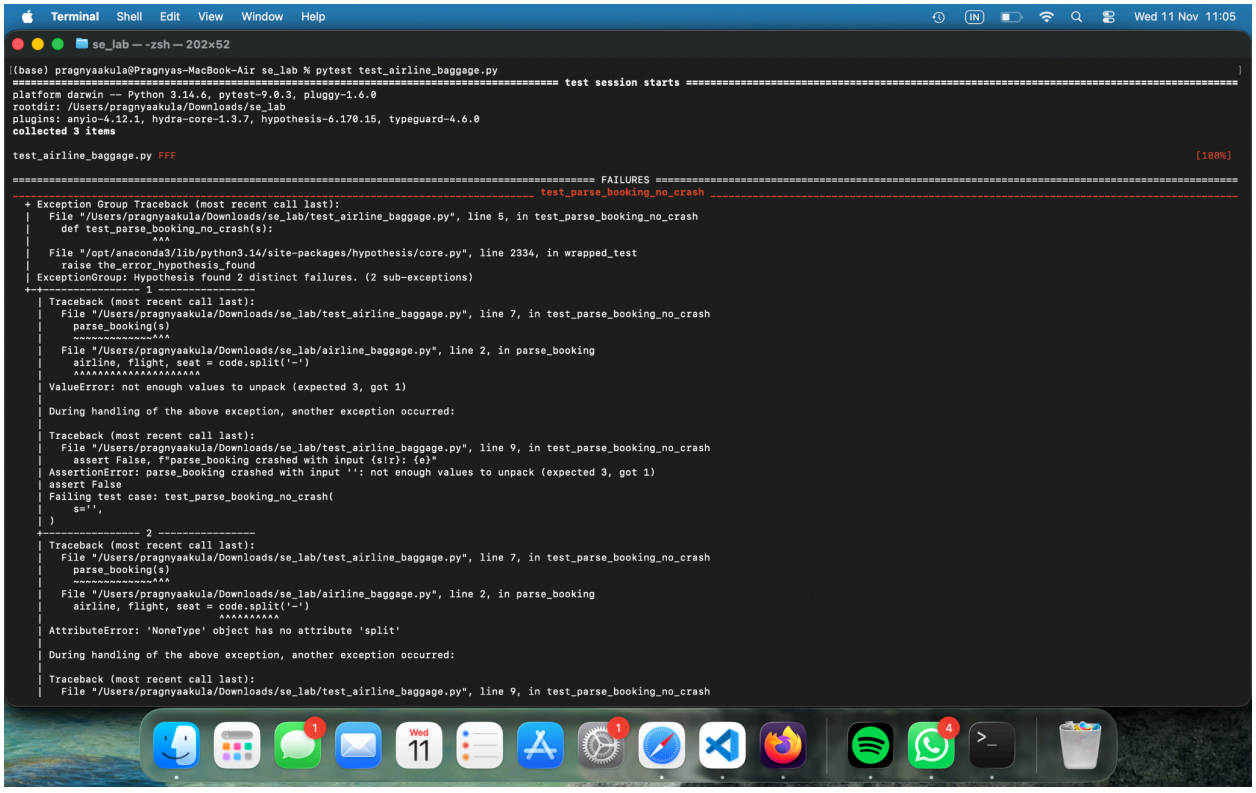
LAB 7: Fuzzing for Input Validation Bugs

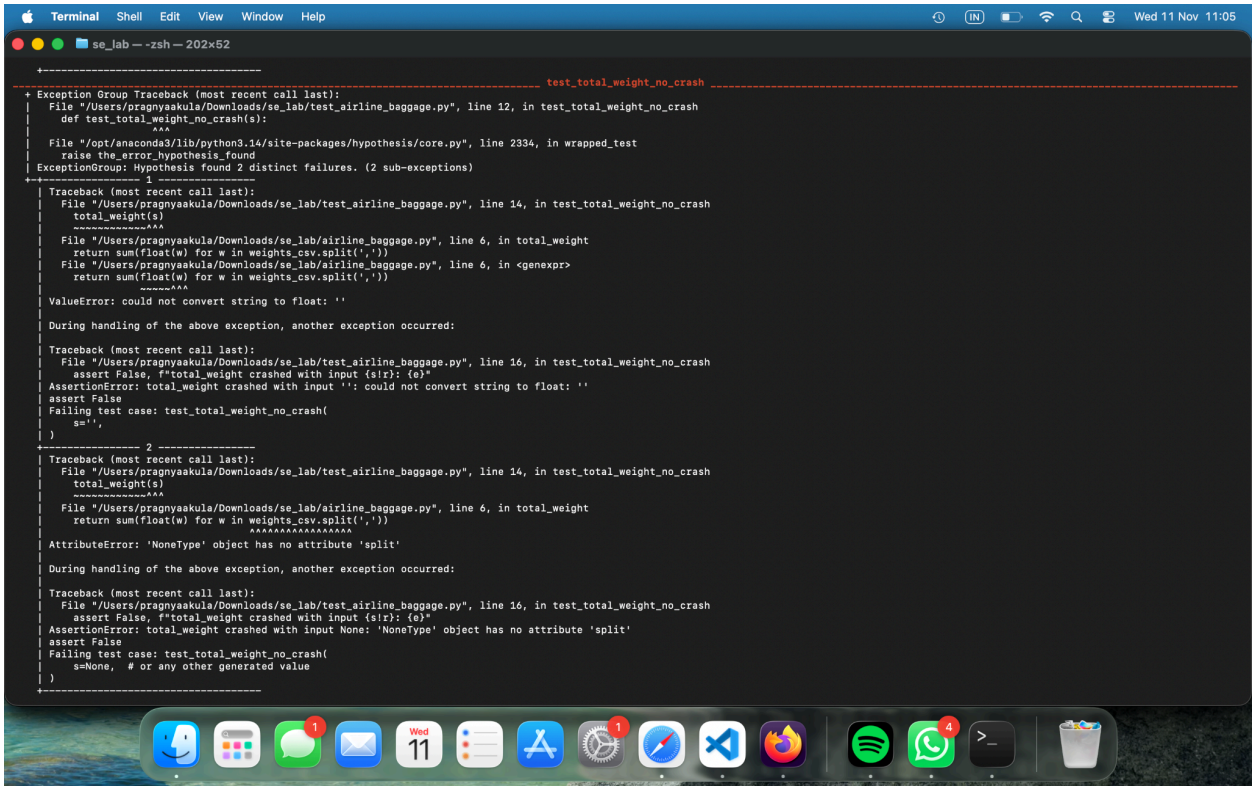
NAME: AKULA PRAGNYA

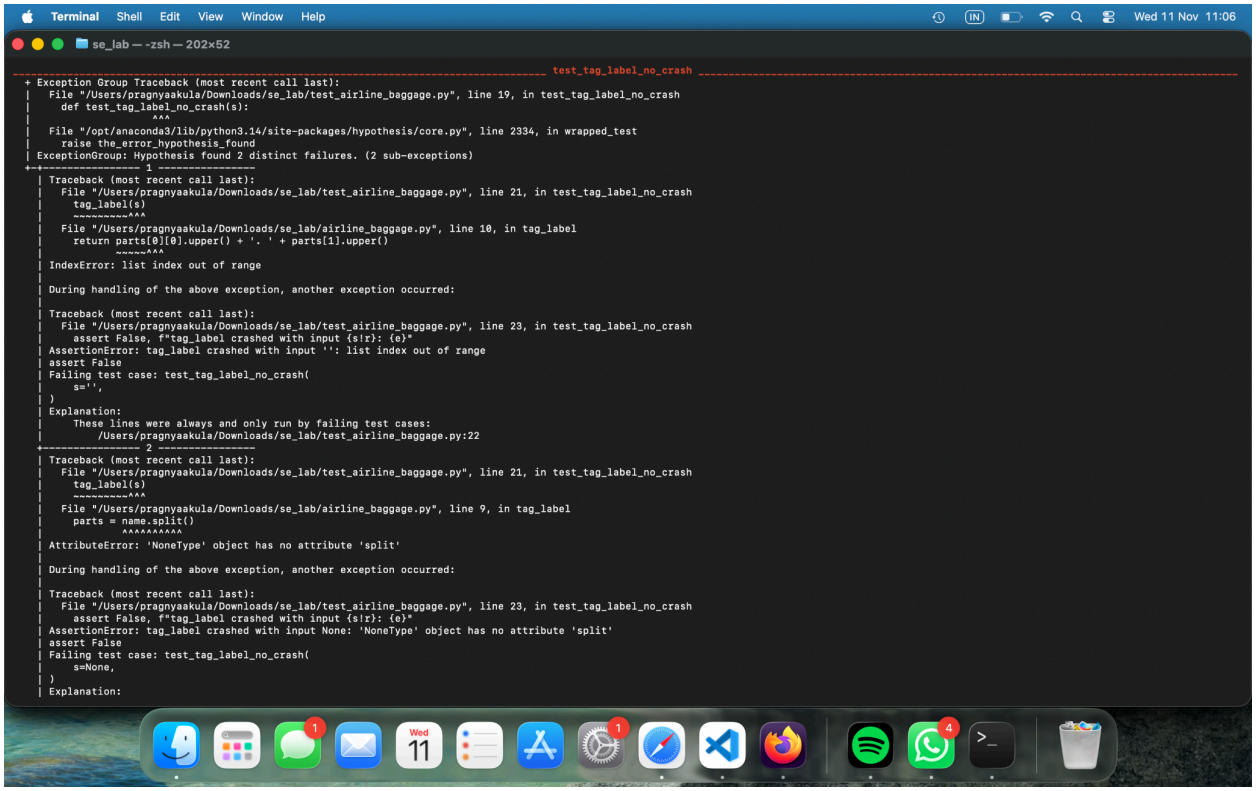
SRN: PES1UG24AM028

SECTION : A

Initial Script & Baseline Interactive Run




Terminal Shell Edit View Window Help
Wed 11 Nov 11:05
se_lab — -zsh — 202×52
test_total_weight_no_crash
+ Exception Group Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 12, in test_total_weight_no_crash
| def test_total_weight_no_crash(s):
| ^^^
| File "/opt/anaconda3/lib/python3.14/site-packages/hypothesis/core.py", line 2334, in wrapped_test
| raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------- 1 ----------
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 14, in test_total_weight_no_crash
| total_weight(s)
| ~~~~~~~~~~~^^^
| File "/Users/pragnyaakula/Downloads/se_lab/airline_baggage.py", line 6, in total_weight
| return sum(float(w) for w in weights_csv.split(','))
| File "/Users/pragnyaakula/Downloads/se_lab/airline_baggage.py", line 6, in <genexpr>
| return sum(float(w) for w in weights_csv.split(','))
| ~~~~~^^^
| ValueError: could not convert string to float: ''
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 16, in test_total_weight_no_crash
| assert False, f"total_weight crashed with input {s!r}: {e}"
| AssertionError: total_weight crashed with input '': could not convert string to float: ''
| assert False
| Failing test case: test_total_weight_no_crash(
| s='',
|)
+---------- 2 ----------
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 14, in test_total_weight_no_crash
| total_weight(s)
| ~~~~~~~~~~~^^^
| File "/Users/pragnyaakula/Downloads/se_lab/airline_baggage.py", line 6, in total_weight
| return sum(float(w) for w in weights_csv.split(','))
| ^^^^^^^^^^^^^^^^
| AttributeError: 'NoneType' object has no attribute 'split'
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 16, in test_total_weight_no_crash
| assert False, f"total_weight crashed with input {s!r}: {e}"
| AssertionError: total_weight crashed with input None: 'NoneType' object has no attribute 'split'
| assert False
| Failing test case: test_total_weight_no_crash(
| s=None, # or any other generated value
|)


Terminal Shell Edit View Window Help
Wed 11 Nov 11:06
se_lab — -zsh — 202×52
test_tag_label_no_crash
+ Exception Group Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 19, in test_tag_label_no_crash
| def test_tag_label_no_crash(s):
| ^^^
| File "/opt/anaconda3/lib/python3.14/site-packages/hypothesis/core.py", line 2334, in wrapped_test
| raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------- 1 ----------
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 21, in test_tag_label_no_crash
| tag_label(s)
| ~~~~~~~~~^^^
| File "/Users/pragnyaakula/Downloads/se_lab/airline_baggage.py", line 10, in tag_label
| return parts[0][0].upper() + '. ' + parts[1].upper()
| ~~~~~^^^
| IndexError: list index out of range
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 23, in test_tag_label_no_crash
| assert False, f"tag_label crashed with input {s!r}: {e}"
| AssertionError: tag_label crashed with input '': list index out of range
| assert False
| Failing test case: test_tag_label_no_crash(
| s='',
|)
| Explanation:
| These lines were always and only run by failing test cases:
| /Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py:22
+---------- 2 ----------
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 21, in test_tag_label_no_crash
| tag_label(s)
| ~~~~~~~~~^^^
| File "/Users/pragnyaakula/Downloads/se_lab/airline_baggage.py", line 9, in tag_label
| parts = name.split()
| ^^^^^^^^^
| AttributeError: 'NoneType' object has no attribute 'split'
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
| File "/Users/pragnyaakula/Downloads/se_lab/test_airline_baggage.py", line 23, in test_tag_label_no_crash
| assert False, f"tag_label crashed with input {s!r}: {e}"
| AssertionError: tag_label crashed with input None: 'NoneType' object has no attribute 'split'
| assert False
| Failing test case: test_tag_label_no_crash(
| s=None,
|)
| Explanation:

```
    +----------------------------------
=========================== short test summary info ============================
FAILED test_airline_baggage.py::test_parse_booking_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_airline_baggage.py::test_total_weight_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_airline_baggage.py::test_tag_label_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 0.48s ===============================
```

Verification Final Execution

```
(base) pragnyaakula@Pragnyas-MacBook-Air se_lab % pytest test_airline_baggage.py
============================= test session starts ==============================
platform darwin -- Python 3.14.6, pytest-9.0.3, pluggy-1.6.0
rootdir: /Users/pragnyaakula/Downloads/se_lab
plugins: anyio-4.12.1, hydra-core-1.3.7, hypothesis-6.170.15, typeguard-4.6.0
collected 3 items

test_airline_baggage.py ...                                              [100%]

============================== 3 passed in 0.95s ===============================
```